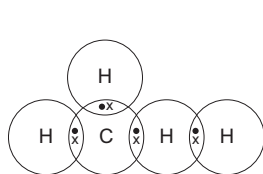
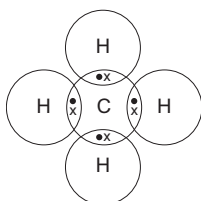
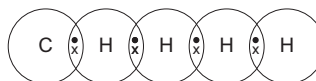
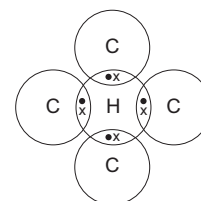


3. (a) Methane gas,  $\text{CH}_4$ , contains the elements carbon and hydrogen.

| Element  | Electronic structure |
|----------|----------------------|
| carbon   | 2,4                  |
| hydrogen | 1                    |

- (i) I. Put a **tick** (✓) in the box below the diagram which shows how the atoms are bonded in one molecule of methane. [1]


☐

☐

☐

☐

- II. **Circle** the name given to the type of bonding found in methane. [1]

**covalent**

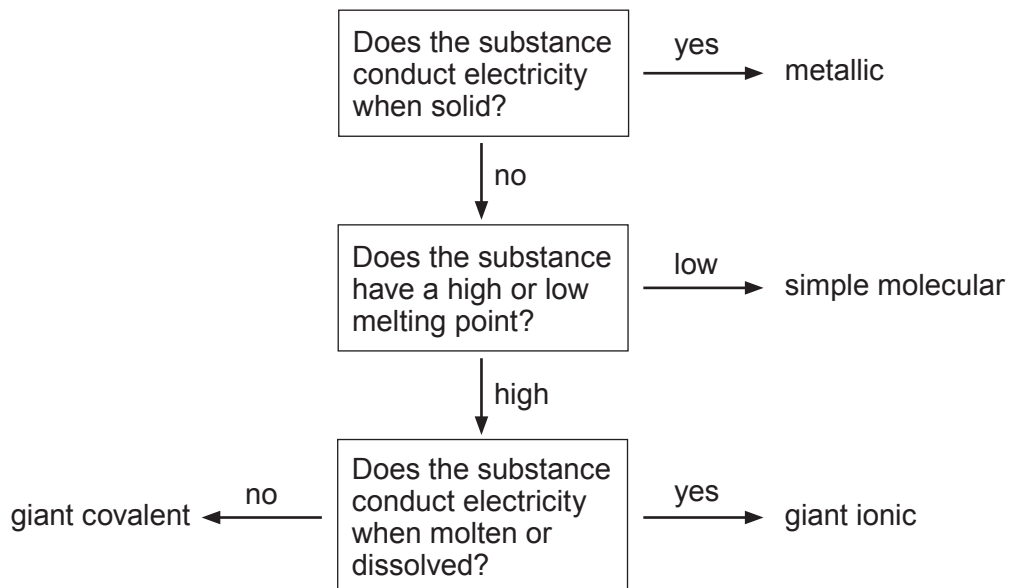
**ionic**

**giant covalent**

**metallic**



- (ii) The flow chart below can be used to identify the type of structure found in different substances.



- I. Use the flow chart to complete the table.

[1]

| Substance | Melting point (°C) | Electrical conductivity      | Type of structure |
|-----------|--------------------|------------------------------|-------------------|
| <b>A</b>  | 2072               | conducts only when molten    | .....             |
| <b>B</b>  | -182               | does not conduct electricity | .....             |
| <b>C</b>  | 1610               | does not conduct electricity | .....             |

- II. Give the **letter** of the substance, **A**, **B** or **C**, that is most likely to be methane. [1]

Substance .....



- (b) Butene,  $C_4H_8$ , also contains the elements carbon and hydrogen. Calculate the percentage by mass of hydrogen in butene. [2]

$$A_r(H) = 1 \quad A_r(C) = 12$$

Percentage = ..... %

Examiner  
only

6

